

Improving E-Commerce Sales & Customer Satisfaction

Brazilian E-Commerce Analysis: Identifying Growth Opportunities & Operational Improvements

99,441

Orders Analyzed

2016–2018 Olist dataset

9

Data Sources

Related CSV datasets joined

26

Final Columns

After cleaning & feature engineering

3

Core Focus Areas

Categories, Delivery, Geography

This analysis surfaces **five key business findings** from 99,441 Brazilian e-commerce orders to guide strategic decisions in logistics, merchandising, and regional growth.

Data & Methodology

Dataset Overview

Source

Brazilian E-Commerce Public Dataset by Olist
(Kaggle)

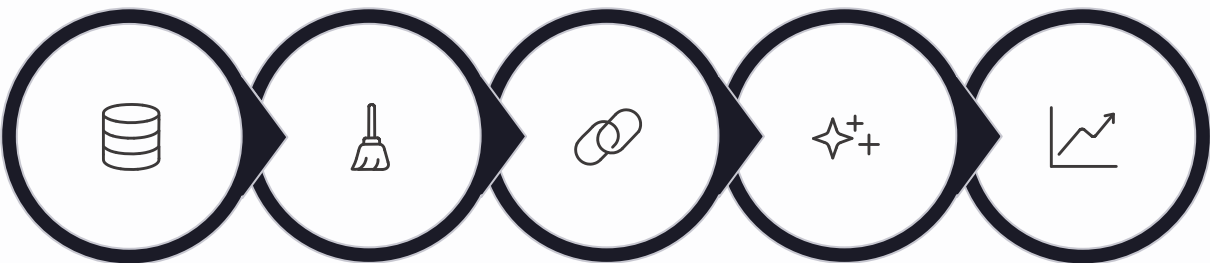
Scope

99,441 orders · 9 CSV datasets · 2016–2018 · 26
engineered columns

Coverage

Complete order, customer, product, and delivery
information per record

Analysis Workflow



Raw Data

Cleaning

Joining

Feature Eng.

EDA

Tools & Technologies

Python

Pandas & NumPy for data processing

Google Colab

Cloud analysis environment

Google Sheets

Data validation

Looker Studio

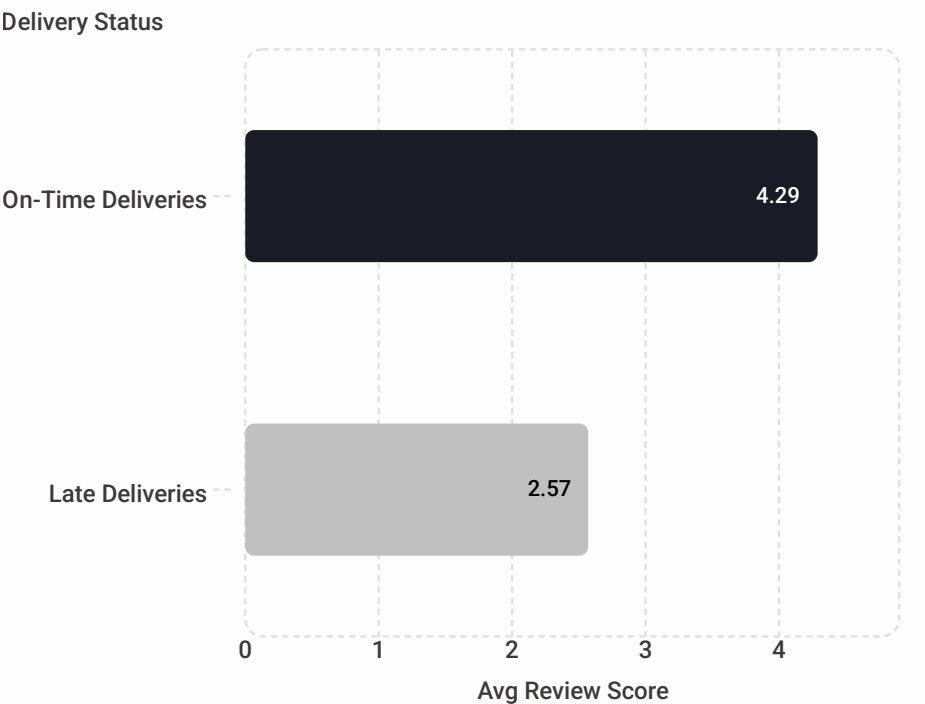
Interactive dashboard

Key Business Findings

Category Concentration & Geographic Distribution

Category Concentration Top 5 product categories contribute 39.74% of total product sales value – a meaningful concentration risk and opportunity signal.	Geographic Concentration São Paulo alone accounts for 41,746 orders (42%). The top 3 states – São Paulo, Rio de Janeiro, and Minas Gerais – represent 66.5% of all orders.
Late Delivery Rate Only 8.11% of delivered orders were late – but this minority is strongly associated with a 40% drop in customer satisfaction scores.	Sales Growth Trend Sales increased substantially during 2017 and remained strong through most of 2018, confirming a growing and active market.

Delivery Performance vs. Customer Satisfaction

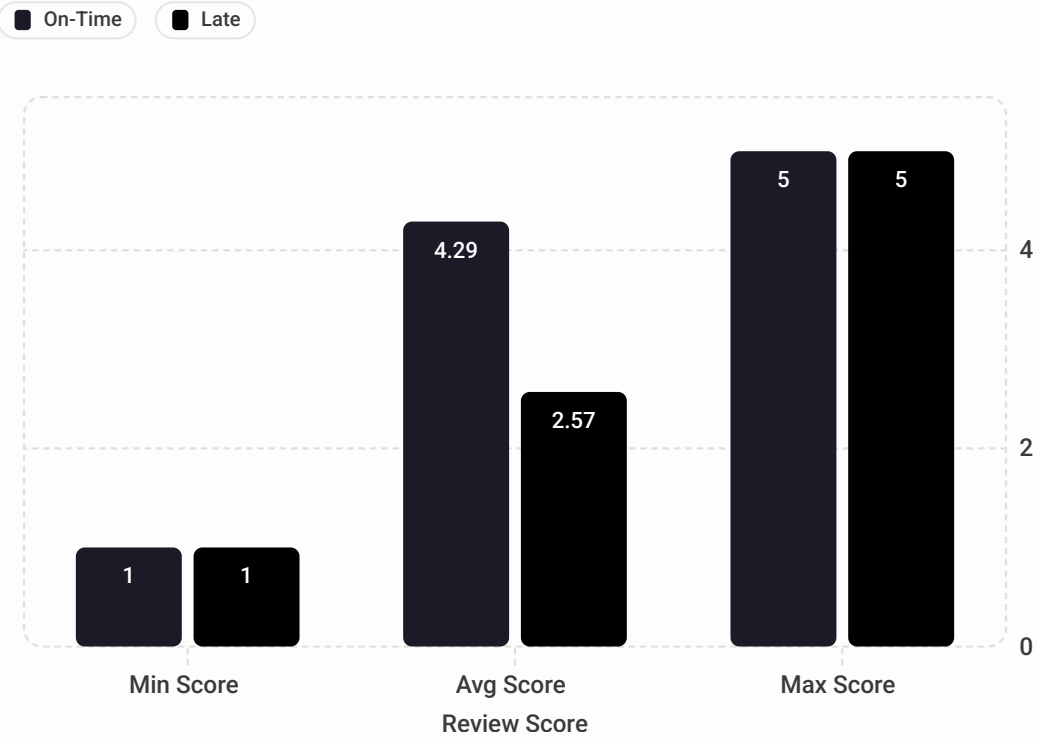


❏ A 1.72-point gap in review scores – **40% lower satisfaction** for late orders. Delivery reliability is a critical customer experience driver.

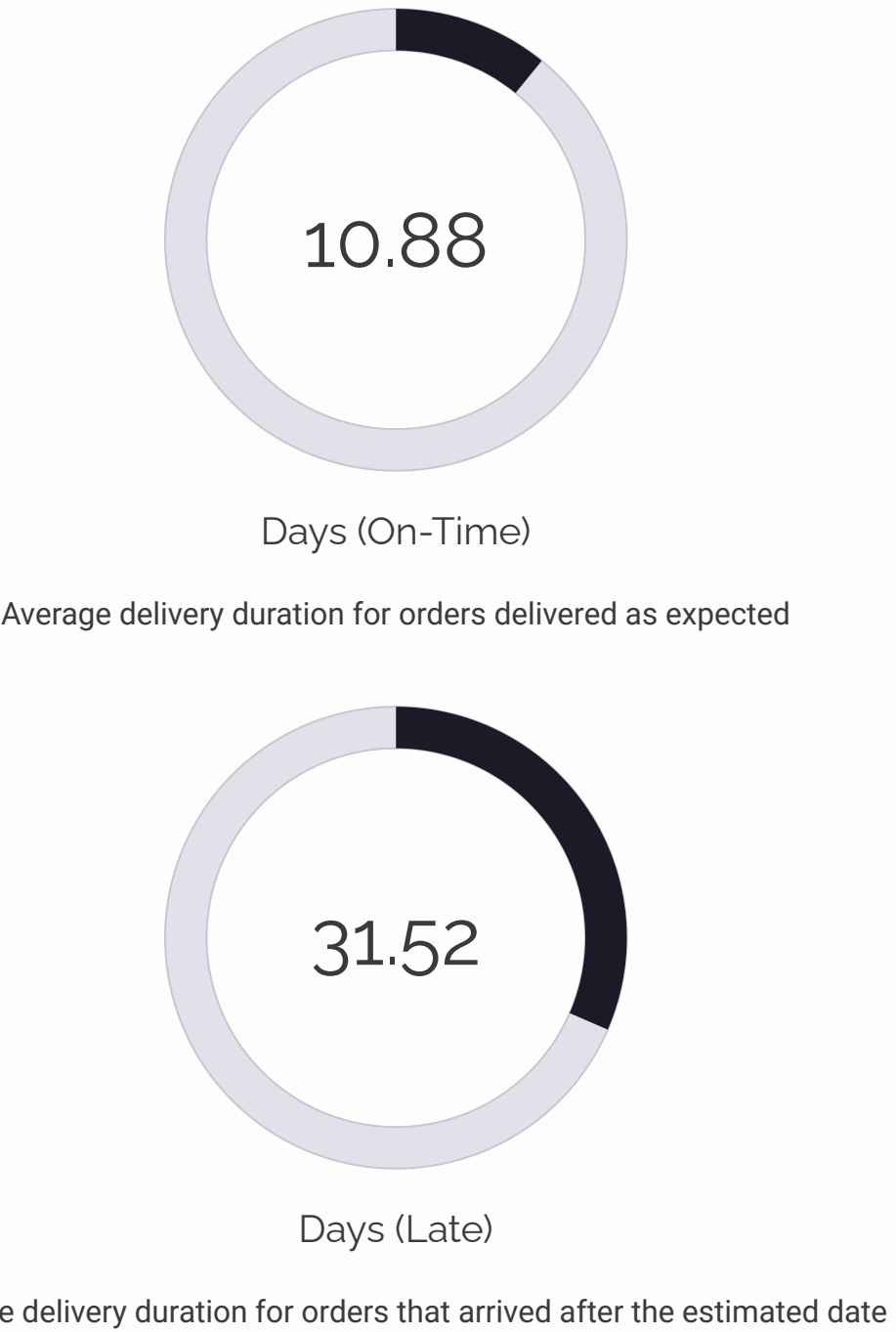
Deep Dive: Delivery Performance & Customer Satisfaction

On-Time Deliveries Avg Review Score: 4.29 / 5.0 Avg Delivery Time: 10.88 days Customer Sentiment: Highly satisfied	Late Deliveries Avg Review Score: 2.57 / 5.0 Avg Delivery Time: 31.52 days Customer Sentiment: Significantly dissatisfied
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Review Score Distribution by Delivery Status



Delivery Time Comparison



📌 *Note: This analysis demonstrates a strong **association** between delivery performance and customer ratings — not causation. Other factors may also influence review scores.*

High-Priority Recommendations

1

HIGH PRIORITY

Reduce Late Deliveries

Action: Identify and address problematic sellers, regions, and product categories with disproportionately high late-delivery rates. Implement seller performance standards and logistics monitoring.

Owner: Logistics & Operations

Success KPIs: Late-delivery percentage · Average review score across all orders

2

HIGH PRIORITY

Prioritize High-Performing Categories

Action: Focus inventory allocation, marketing spend, and seller recruitment on strong-performing categories — particularly Health & Beauty and Watches & Gifts, which anchor the top 5 contributing 39.74% of sales value.

Owner: Merchandising / Marketing

Success KPIs: Product sales value · Category contribution percentage

3

MEDIUM PRIORITY

Optimize Key Geographic Markets

Action: Strengthen logistics infrastructure and tailor regional strategies in São Paulo, Rio de Janeiro, and Minas Gerais — which together represent 66.5% of all orders. Prioritize on-time delivery performance in these high-volume markets.

Owner: Regional Operations / Marketing

Success KPIs: Sales volume · Order count · On-time delivery rate by state

Expected Business Impact



Operational Impact

- Fewer late deliveries and improved logistics efficiency
- More reliable and predictable delivery performance
- Reduced operational friction and customer service costs



Customer Impact

- Higher average customer review scores
- Improved customer satisfaction and retention rates
- Better word-of-mouth and repeat purchase frequency



Commercial Impact

- Better inventory allocation to high-performing categories
- Stronger sales growth in priority product segments
- More effective regional marketing and seller strategies
- Improved competitive positioning in key markets

Target Success Metrics

Late-Delivery Rate

Reduce from current baseline of **8.11%** through targeted seller and logistics interventions

Avg Review Score

Increase overall customer review score across all order types as delivery reliability improves

Category Sales Share

Grow sales contribution from priority categories above the current **39.74%** concentration

On-Time Delivery by State

Improve on-time delivery performance specifically in SP, RJ, and MG – 66.5% of all orders

Limitations & Next Steps

Analysis Limitations

Historical Data

Dataset covers 2016–2018 and may not reflect current customer behavior or market conditions.

Missing Business Data

No cost, profit-margin, marketing-spend, or customer-acquisition data was available, limiting ROI analysis.

Data Completeness

Some orders have missing sales, delivery, or review information that could affect aggregate results.

Extreme Values

Extreme delivery durations were retained as they may represent genuine operational edge cases.

Association vs. Causation

All delivery-satisfaction findings reflect statistical association only — causation has not been established.

Recommended Next Steps

01

Root-Cause Investigation

Drill down into late deliveries by seller, region, and product category to identify systemic causes and prioritize interventions.

02

Profitability Data Collection

Collect cost, margin, and customer-acquisition data to enable deeper ROI analysis and resource prioritization.

03

Current-Period Monitoring

Track live customer satisfaction and delivery performance metrics to validate findings against current conditions.

04

Broader Satisfaction Drivers

Perform deeper analysis to identify additional factors influencing review scores beyond delivery performance.

05

Predictive Modelling

Develop models to forecast delivery delays and customer churn risk, enabling proactive operational responses.